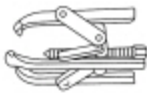


# Cylinder Head



## Special Tool(s)

|                                                                                                |                                                               |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <br>ST1184-A   | 3 Jaw Puller<br>303-D121 or equivalent                        |
| <br>ST2132-A   | Adapter Set, Step Plate<br>205-DS011 or equivalent            |
| <br>ST1377-A   | Lifting Bracket, Engine<br>303-F047 (014-00073) or equivalent |
| <br>ST1385-A | Remover, Oil Seal<br>303-409 (T92C-6700-CH)                   |
| <br>ST3110-A | Support Brackets, Engine<br>303-1507 or equivalent            |

## Material

| Item                                            | Specification |
|-------------------------------------------------|---------------|
| Motorcraft® Metal Surface Prep Wipes<br>ZC-31-B | —             |
| Motorcraft® Silicone Gasket Remover<br>ZC-30-A  | —             |

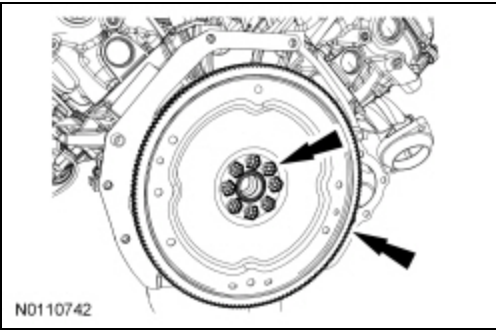
**NOTICE:** During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

**NOTE:** The flexplate and spacer plate must be removed before mounting the engine on the engine stand.

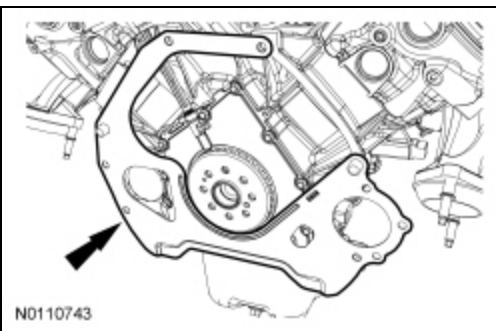
**NOTE:** For additional information, refer to the exploded view under the Assembly procedure in this section.

## All cylinder heads

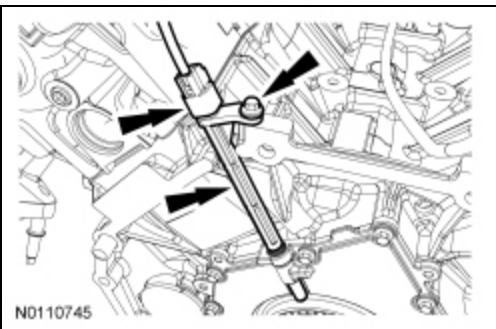
1. Remove the engine. For additional information, refer to Engine in this section.
2. Remove the 8 bolts and the flexplate.
  - Discard the 8 flexplate bolts.



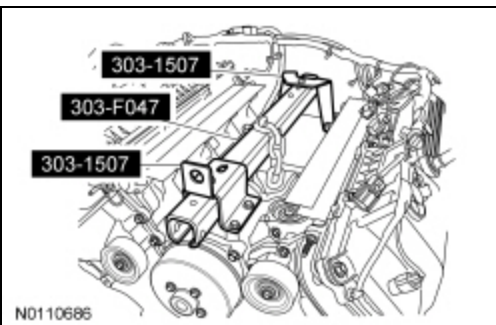
3. Remove the spacer plate.



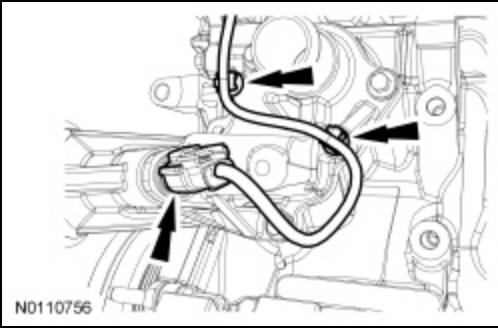
4. Disconnect the electrical connector and remove the bolt and the Crankshaft Position (CKP) sensor.



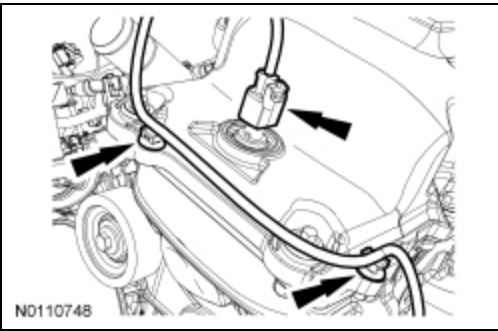
5. Mount the engine on a work stand.
6. Remove the Engine Lifting Brackets.



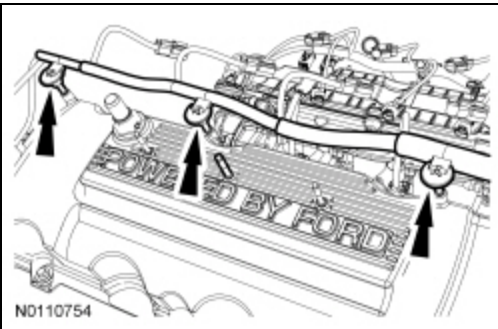
7. Disconnect the Engine Oil Pressure (EOP) switch electrical connector.



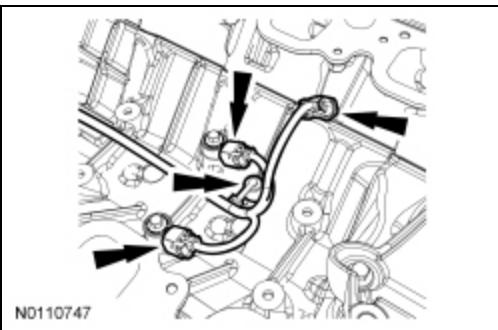
8. Disconnect the LH Variable Camshaft Timing (VCT) variable force solenoid electrical connector and the 2 wiring harness retainers.



9. Disconnect the 3 engine wiring harness position retainers from the LH valve cover.



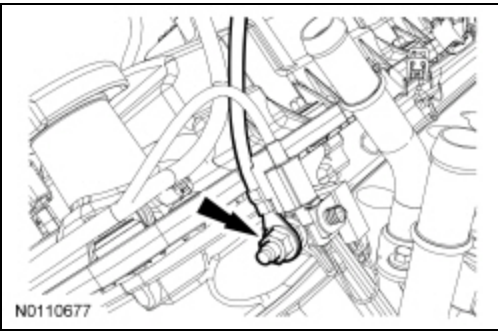
10. Disconnect the wiring harness retainer, the Knock Sensor (KS) and Cylinder Head Temperature (CHT) sensor electrical connector.



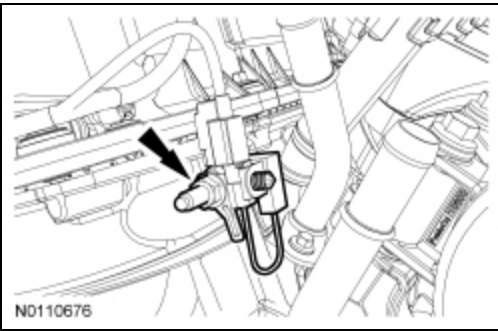
11. Disconnect the LH and RH Camshaft Position (CMP) sensor electrical connectors.



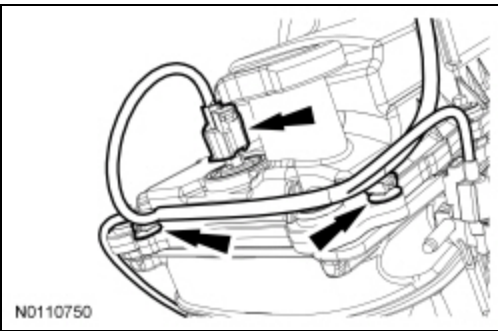
12. Remove the nut and the wiring harness ground cable.



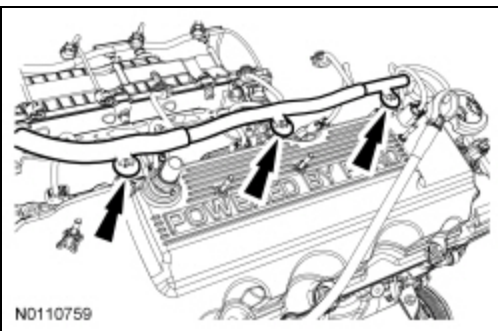
13. Remove the nut and the radio interference capacitor.



14. Disconnect the RH VCT variable force solenoid electrical connector and the 2 wiring harness retainers.

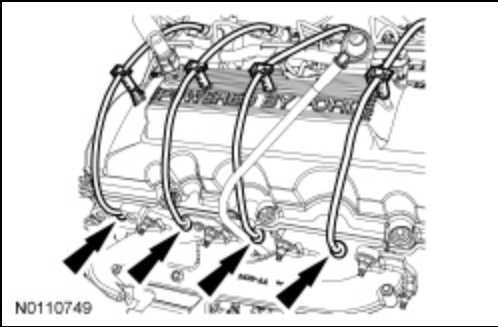


15. Disconnect the 3 engine wiring harness position retainers from the RH valve cover and remove the engine wiring harness.

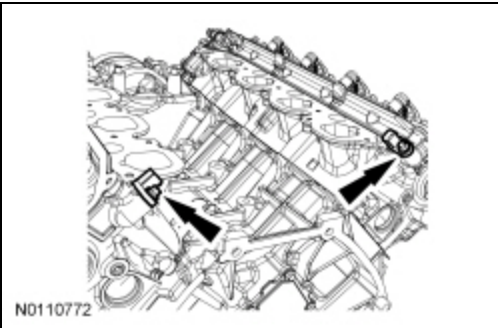


16. **NOTE:** *RH shown, LH similar.*

Disconnect the 8 ignition wires from the valve cover retainers and the lower spark plugs.

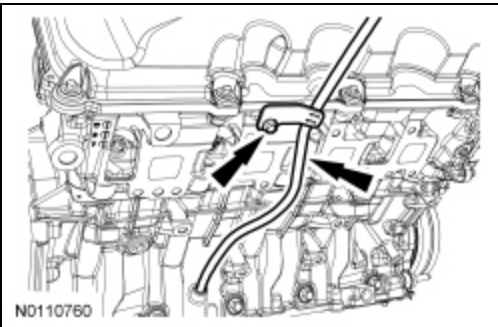


17. Remove the 2 bolts and the LH and RH CMP sensors.



18. Remove the bolt and the oil level indicator tube.

- Discard the O-ring seal.



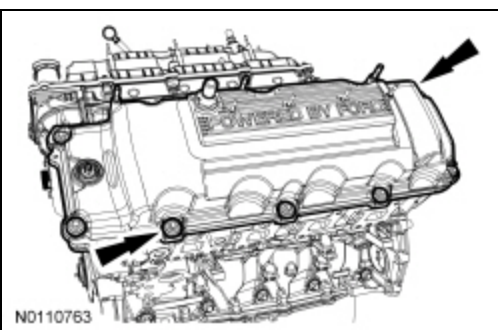
19. **NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

**NOTICE:** Remove the valve cover carefully, or the Variable Camshaft Timing (VCT) variable force solenoid may be damaged.

**NOTE:** The bolts are part of the valve cover and should not be removed.

Loosen the 10 bolts and remove the LH valve cover.

- Clean the valve cover mating surface of the cylinder head with silicone gasket remover and metal surface prep. Follow the directions on the packaging.
- Discard the valve cover gasket. Clean the valve cover gasket groove with soap and water or a suitable solvent.



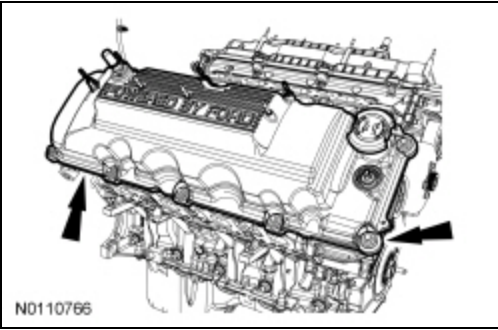
20. **NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

**NOTICE:** Remove the valve cover carefully, or the Variable Camshaft Timing (VCT) variable force solenoid may be damaged.

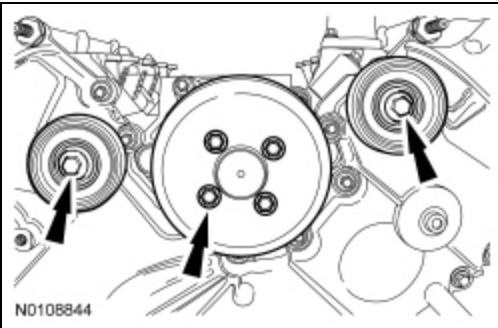
**NOTE:** The bolts are part of the valve cover and should not be removed.

Loosen the 10 bolts and remove the RH valve cover.

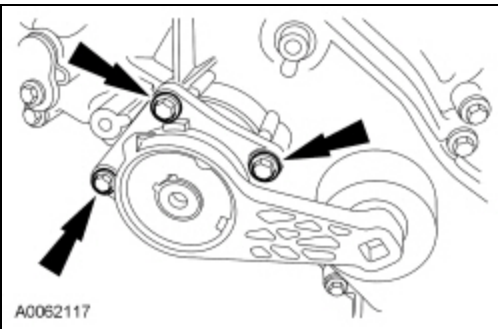
- Clean the valve cover mating surface of the cylinder head with silicone gasket remover and metal surface prep. Follow the directions on the packaging.
- Discard the valve cover gasket. Clean the valve cover gasket groove with soap and water or a suitable solvent.



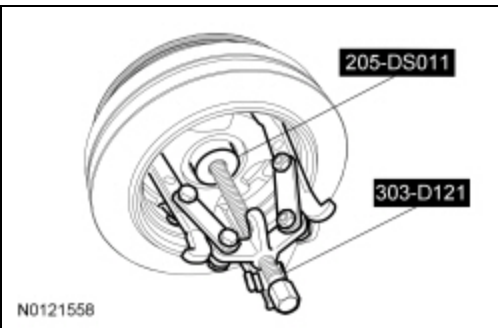
21. Remove the 6 bolts, the coolant pump pulley and the 2 accessory drive belt idler pulleys.



22. Remove the 3 bolts and the accessory drive belt tensioner.



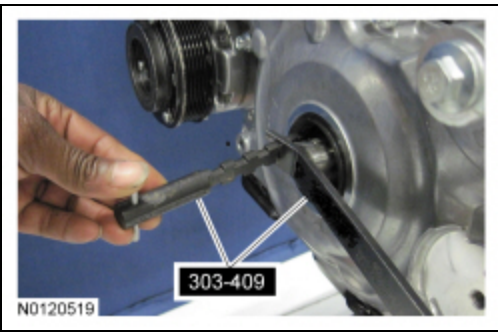
23. Using the 3 Jaw Puller and Step Plate, remove the crankshaft pulley.



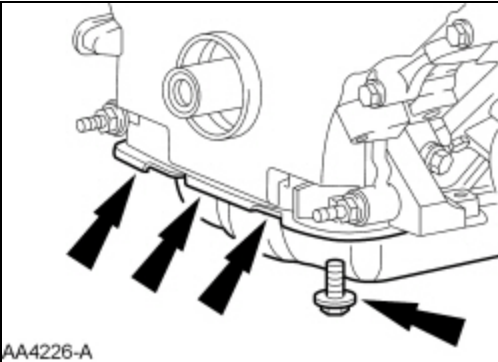


24. **NOTICE:** Use care not to damage the engine front cover or the crankshaft when removing the seal.

Using the Oil Seal Remover, remove the crankshaft front oil seal.

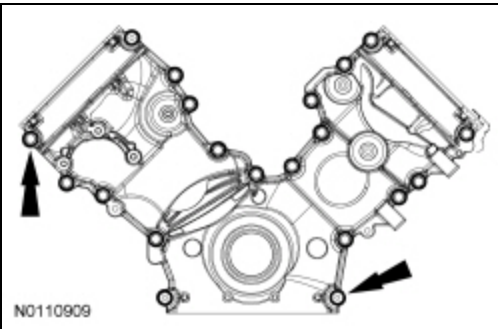


25. Remove the front 4 oil pan bolts.



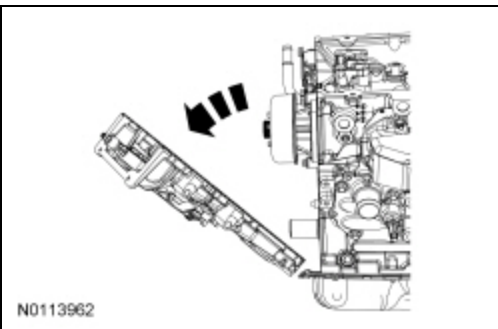
26. **NOTE:** *Correct fastener location is essential for the assembly procedure. Record fastener location.*

Remove the 17 bolts and the 3 studs.



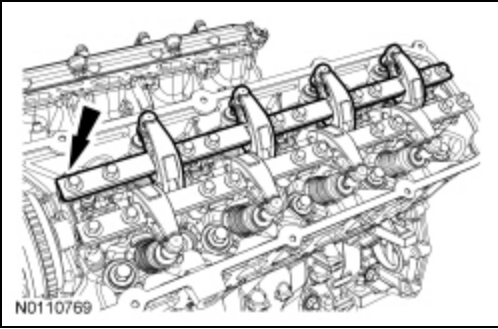
27. Remove the engine front cover from the front cover-to-cylinder block dowel.

- Remove and discard the engine front cover gaskets.



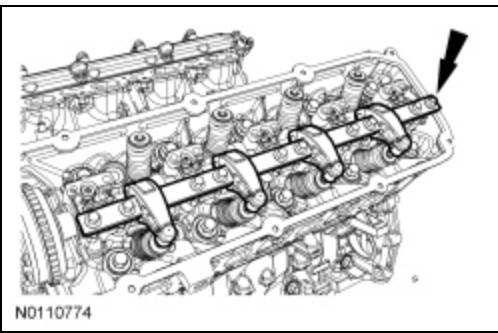
28. Rotate the engine clockwise until the No. 1 intake valve is open.

29. Remove the 10 bolts and the LH intake rocker arm shaft assembly.



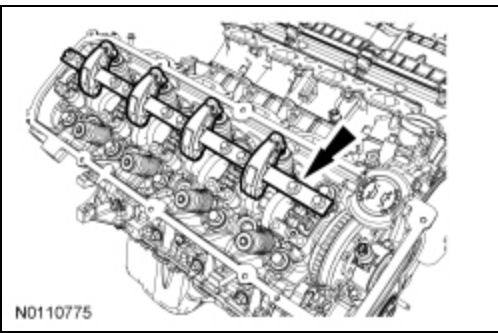
30. Rotate the engine clockwise until the No. 1 exhaust valve is open.

31. Remove the 10 bolts and the LH exhaust rocker arm shaft assembly.



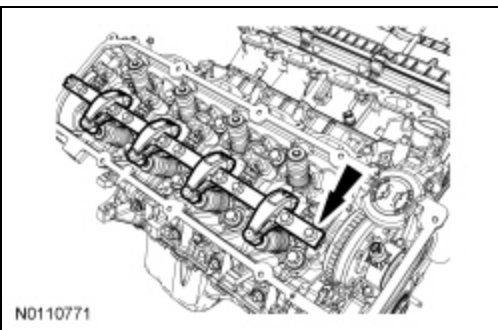
32. Rotate the engine clockwise until the No. 1 intake valve is closed.

33. Remove the 10 bolts and the RH intake rocker arm shaft assembly.



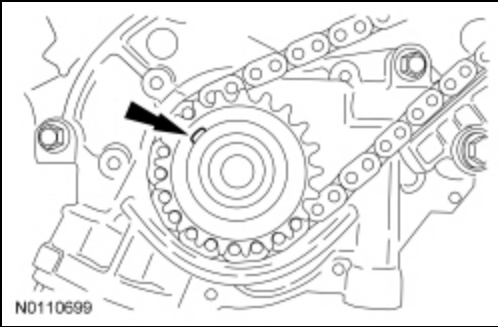
34. Rotate the engine clockwise until the No. 1 exhaust valve is closed.

35. Remove the 10 bolts and the RH exhaust rocker arm shaft assembly.

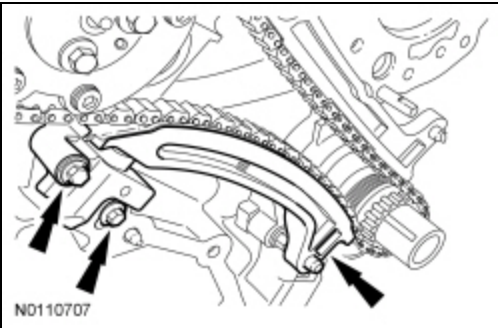


36. Position the crankshaft keyway at the 11 o'clock position.

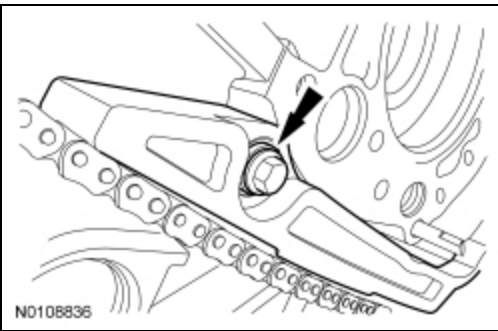




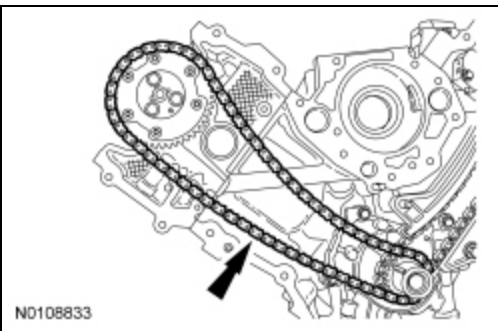
37. Remove the 2 bolts, the RH timing chain tensioner and tensioner arm.



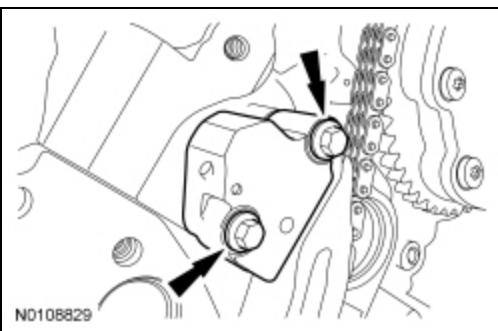
38. Remove the bolt and the RH timing chain guide.



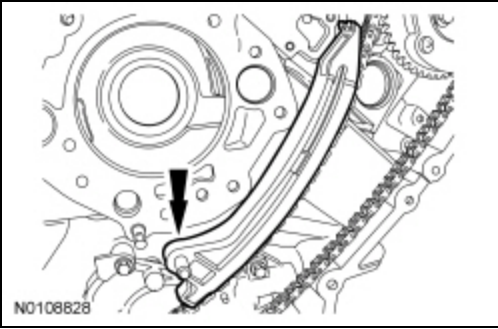
39. Remove the RH timing chain.



40. Remove the 2 bolts and the LH timing chain tensioner.



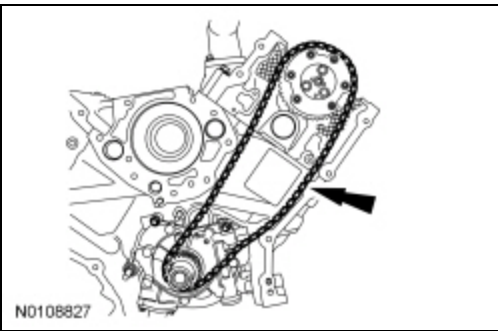
41. Remove the LH timing chain tensioner arm.



42. Remove the bolt and the LH timing chain guide.

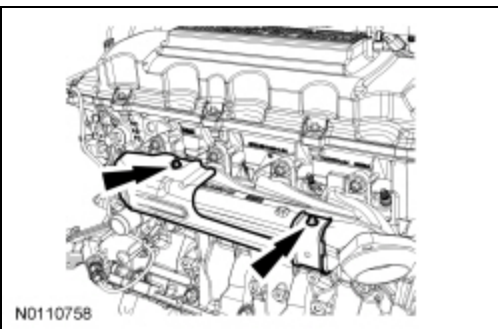


43. Remove the LH timing chain.



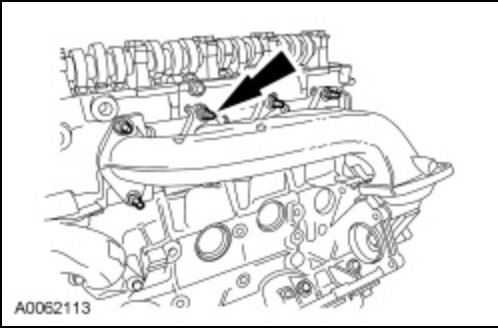
## LH cylinder head

44. Remove the 2 bolts and the LH exhaust manifold heat shield.



45. Remove the 8 nuts, the 8 studs, the LH exhaust manifold and gaskets.

- Discard the 8 nuts and the 8 studs.
- Discard the 2 gasket.
- Inspect the exhaust manifold. For additional information, refer to Section 303-00.



46. **NOTICE:** The cylinder head must be cool before removing it from the engine. Cylinder head warpage may result if a warm or hot cylinder head is removed.

**NOTICE:** Place clean shop towels over exposed engine cavities. Carefully remove the towels so foreign material is not dropped into the engine.

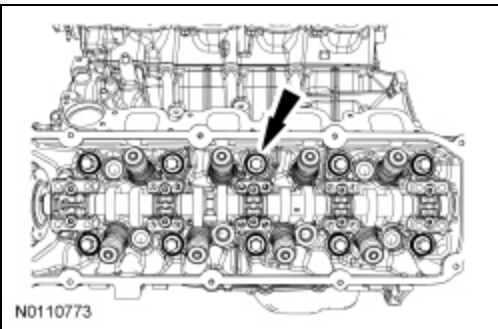
**NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges that make leak paths. Use a plastic scraping tool to remove all traces of the head gasket.

**NOTICE:** Aluminum surfaces are soft and can be scratched easily. Never place the cylinder head gasket surface, unprotected, on a bench surface, or the cylinder head may be damaged.

**NOTE:** The cylinder head bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.

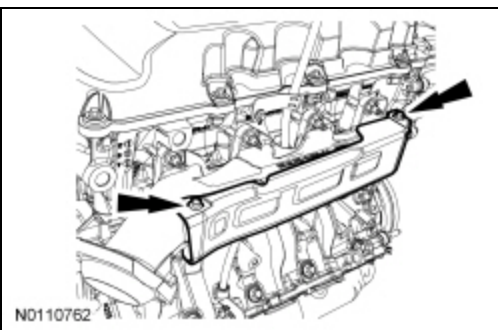
Remove the 11 bolts and the LH cylinder head.

- Discard the cylinder head gasket.
- Discard the 11 cylinder head bolts.

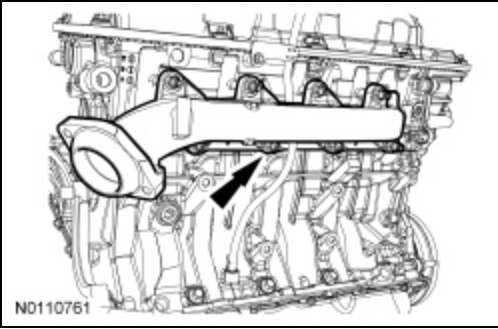


### RH cylinder head

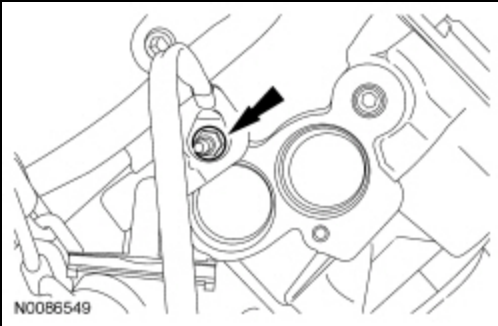
47. Remove the 2 bolts and the RH exhaust manifold heat shield.



48. Remove the 8 nuts, the 8 studs, the RH exhaust manifold and gaskets.
- Discard the 8 nuts and the 8 studs.
  - Discard the 2 gasket.
  - Inspect the exhaust manifold. For additional information, refer to Section 303-00.



49. Remove the nut and the ground strap.



50. **NOTICE:** The cylinder head must be cool before removing it from the engine. Cylinder head warpage may result if a warm or hot cylinder head is removed.

**NOTICE:** Place clean shop towels over exposed engine cavities. Carefully remove the towels so foreign material is not dropped into the engine.

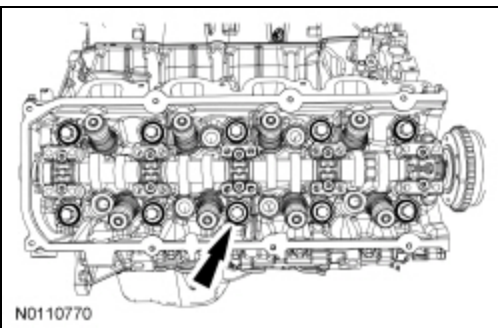
**NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges that make leak paths. Use a plastic scraping tool to remove all traces of the head gasket.

**NOTICE:** Aluminum surfaces are soft and can be scratched easily. Never place the cylinder head gasket surface, unprotected, on a bench surface, or the cylinder head may be damaged.

**NOTE:** The cylinder head bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.

Remove the 10 bolts and the RH cylinder head.

- Discard the cylinder head gasket.
- Discard the 11 cylinder head bolts.



### All cylinder heads

51. **NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges that make leak paths. Use a plastic scraping tool to remove all traces of the head gasket.

**NOTE:** *Observe all warnings or cautions and follow all application directions contained on the packaging of the silicone gasket remover and the metal surface prep.*

**NOTE:** *If there is no residual gasket material present, metal surface prep can be used to clean and prepare the surfaces.*

Clean the cylinder head-to-cylinder block mating surfaces of both the cylinder head and the cylinder block in the following sequence.

1. Remove any large deposits of silicone or gasket material with a plastic scraper.
2. Apply silicone gasket remover, following package directions and allow to set for several minutes.
3. Remove the silicone gasket remover with a plastic scraper. A second application of silicone gasket remover may be required if residual traces of silicone or gasket material remain.
4. Apply metal surface prep, following package directions, to remove any remaining traces of oil or coolant and to prepare the surfaces to bond with the new gasket. Do not attempt to make the metal shiny. Some staining of the metal surfaces is normal.

52. Support the cylinder heads on a bench with the head gasket side up. Check the cylinder head distortion and the cylinder block distortion, paying particular attention to the oil pressure feed area. For additional information, refer to Section 303-00.